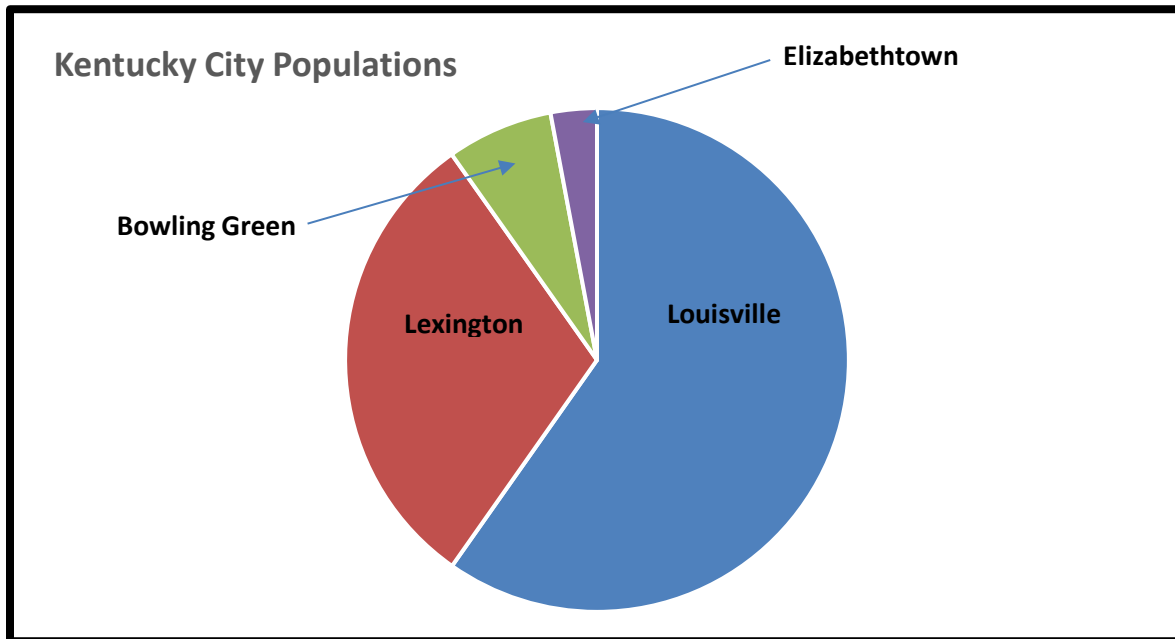




## Mathematics Written Assessment 2023 Elementary Region Governor's Cup

### Student Instructions:

- For each question, choose the BEST answer.
- What is the value of  $20^2 \div 10 + 2$ ?  
A. 4  
B. 33.3333  
C. 420  
D. 42
  - Which set of numbers is arranged from least to greatest?  
A.  $2027.0 < 202.70 < 20.270 < 2.0270$   
B.  $2.0270 < 20.270 < 202.70 < 202.0$   
C. 500, 700, 400, 300  
D. 23,25,29,31
  - What is the value of the underlined digit: 19602.053?  
A. 9 hundred  
B. 9 hundredths  
C. 9 thousand  
D. 9 thousandths

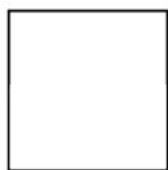


Use the pie chart above to answer questions 4, 5, and 6.

4. Which city has approximately one-fourth of the total population?
- A. Bowling Green
  - B. Elizabethtown
  - C. Lexington
  - D. Louisville
5. The total population of the four cities is 1,059,200. What is the approximate population of the city identified in question number 4?
- A. 264,800
  - B. 529,600
  - C. 653,066
  - D. It is not possible to determine from the given information.
6. Which of the four cities has the smallest population?
- A. Bowling Green
  - B. Elizabethtown
  - C. Lexington
  - D. Louisville
7. How do you write the number 700,757 in words?
- A. Seven hundred seventy thousand fifty-seven
  - B. Seven hundred thousand seven hundred fifty-seven
  - C. Seven thousand seven hundred fifty-seven
  - D. Seven hundred seven five seven
8. What property is shown by the following values?  
If  $A = B$  and  $B = 2,235$ , then  $A = 2,235$
- A. Distributive Property
  - B. Identity Property
  - C. Transitive Property
  - D. Commutative Property
9. Determine the least common multiple of 7, 13, and 169.
- A. 2,197
  - B. 91
  - C. 169
  - D. 1,183
10. In Governor's Cup, there are 30 questions on the math written assessment. Harrison has 35 minutes to complete the exam. How many minutes, rounded to hundredths, should he spend on each question to ensure he finishes all questions?
- A. 1.16 second
  - B. 1 minute 16 seconds
  - C. 1.17 minutes
  - D. 0.85714 minutes

11. Convert  $\frac{7}{16}$  to a decimal, rounded to the thousandth place.
- A. 0.4375
  - B. 0.438
  - C. 0.437
  - D. 0.43
12. The Philadelphia Eagles have a record of 613 wins, 630 losses, and 27 ties in its 90 seasons as a team. Approximately what percent of their total games ended in ties?
- A. 2.1%
  - B. 37%
  - C. 21%
  - D. 3.7%
13. On a map of the United States, one inch equals 557 miles. Kelly measures the distance between Frankfort, Kentucky, to Los Angeles, California. She finds that the distance is 3.85 inches. What is the distance from Frankfort to Los Angeles?
- A. 214,445 miles
  - B. 214.445 miles
  - C. 2144.45 miles
  - D. 144.675 miles
14. February has 29 days in a leap year. How many minutes long is a leap year month?
- A. 104,400
  - B. 1,740
  - C. 696
  - D. 41,760
15. Right triangle ABC is similar to right triangle DEF. Which statement is TRUE about the two triangles?
- A. Triangle ABC is the same size as triangle DEF.
  - B. Angle A is a right angle, so angle F must also be a right angle.
  - C. The angles in triangles ABC and DEF are congruent.
  - D. Both B and C are true.
16. A 2.5-gallon aquarium measures 12 inches by 6 inches by 8 inches. How much water can the aquarium hold when it is only half full?
- A. 576 cubic inches
  - B. 288 cubic inches
  - C. 576 square inches
  - D. 288 square inches
17. Simplify the following numerical expression completely:  $(5-3) (2^3 + 14) (8-8)$
- A. 0
  - B.  $(2) (8 + 14) (0)$
  - C. 2220
  - D. 30
18. What is the value of the algebraic expression if the value of x is 4 and the value of y is 12?
- $$(x + y)^2 - 36/6$$
- A. 36.6667
  - B. 256
  - C. 250
  - D. Impossible to determine from the given information.

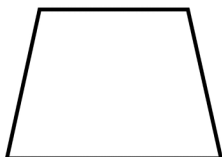
19. Which of the following quadrilaterals has the fewest lines of symmetry?



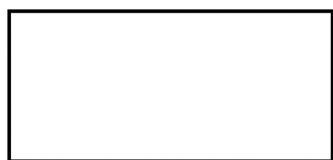
A. square



B. parallelogram



C. isosceles trapezoid



D. rectangle

20.



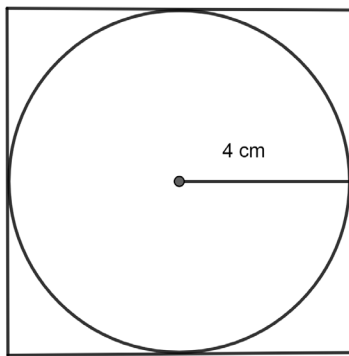
There is one green marble, one white marble, one yellow marble, one red marble, and one blue marble. If you place the marbles in a line, how many possible ways can you arrange the marbles?

- A. 5 ways
- B. 120 ways
- C. 100 ways
- D.  $\frac{1}{120}$  ways

21. Which of the following numbers is not divisible by 3 AND 6?

- A. 18
- B. 36
- C. 54
- D. 33

22. A chocolate cake mix requires  $\frac{2}{3}$  cup of milk. Frankie uses 8 boxes of mix to make desserts for the silent auction. How many cups of milk will she need to make the cakes?
- 8 cups
  - 5 cups
  - $5\frac{1}{3}$  cups
  - $8\frac{1}{3}$  cups
23. There are 27 students in Mr. Hampton's fifth-period class. Twelve of the students have brown hair. Six have blonde hair, and nine have red hair. What percent of the class has red hair?
- 33.3%
  - $\frac{1}{3}\%$
  - 22.2%
  - 44.4%



24. The illustration to the right shows a circle with a radius of 4 centimeters inscribed in a square. What is the perimeter of the square?
- 8 cm.
  - 16 cm.
  - $32\pi$  cm.
  - 32 cm.
25. A spinner is divided into equal sectors containing multiples of ten. The numbers are in no particular order, but the spinner has 4 number tens, 2 number 30s, 6 number 50s, and 3 number 60s. On which number should the spinner land 20% of the time?
- 10
  - 30
  - 50
  - 60
26. There are 24 Chromebooks in the charging station. Mrs. Davis's class borrows  $\frac{1}{4}$  of the devices, and Mr. Hanover's class borrows  $\frac{1}{2}$  of the remaining devices. How many Chromebooks are left in the charging station?
- 12
  - 8
  - 9
  - 0
27. Find the missing number in the pattern shown here: 5, 7, 10, 14, \_\_\_\_\_, 25, 32
- 19
  - 18
  - 17
  - 16

28. Replace the question marks with operational symbols in order from left to right to create a numerical expression that equals 0.

$$75 ? 3 ? 6 ? 31$$

- A.  $- , - , +$
  - B.  $/ , + , -$
  - C.  $> , > , >$
  - D.  $\sqrt{\phantom{x}} , + , -$
29. This statistical term is defined as the value that has a higher frequency in a given set of values. What is the value in a data set that appears most often?
- A. Mean
  - B. Median
  - C. Mode
  - D. Frequency
30. What is the range of this data set?

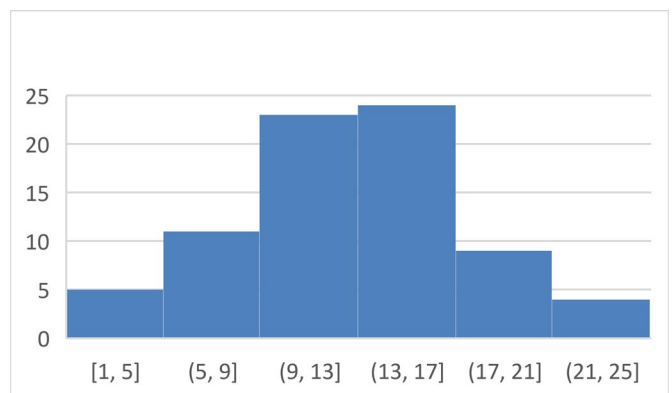
$$23, 76, 15, 93, 2, 112, 15$$

- A. 15
- B. 48
- C. 2
- D. 110

### TIEBREAKERS

31. Kelly built a function table such that the output of each part of the table is six less than double the input. If the input is represented as  $x$  and the output is represented as  $y$ , which equation is an accurate depiction of the function rule?
- A.  $x + 6 = y$
  - B.  $y = 2x - 6$
  - C.  $x = 2y - 6$
  - D.  $y + 6 = x$
32. You can buy four sizes of cereal at the market. An 18-ounce box costs \$2.88. A 13-ounce package costs \$2.60. A 14-ounce box costs \$2.10, and a 2-pound box costs \$3.20. Which box of cereal is the better buy based on unit cost?
- A. 18-ounce box
  - B. 13-ounce box
  - C. 14-ounce box
  - D. 2-pound box

33. The graph at the right is an example of this type of graph. It groups numbers into ranges, and each bar's height shows how many fall into each range. Identify the type of graph that might be used to record the change in a baby's weight over time.
- A. Histogram
  - B. Bar Graph
  - C. Box-and-whisker
  - D. Waterfall

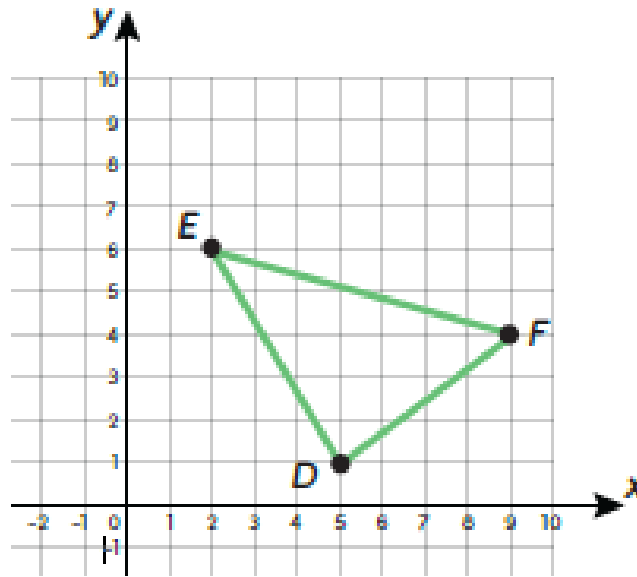


34. If you flip a coin 2 times, what is the probability that you will get no tails?

- A.  $\frac{1}{2}$
- B.  $\frac{2}{3}$
- C.  $\frac{1}{4}$
- D.  $\frac{3}{2}$

35. Look at the coordinate plane shown below. What are the coordinates for Point D of the triangle?

- A. (0,0)
- B. (9,4)
- C. (5,1)
- D. (1,5)



36. The dimensions of a toolbox are 16 inches by 10 inches by 6 inches. What is the volume of the toolbox?

- A. 960 cubic inches
- B. 960 square inches
- C. 160 cubic inches
- D. 60 square inches

# **Mathematics Written Assessment**

## **2023 Elementary Region Governor's Cup**

1. D
2. D
3. C
4. C
5. A
6. B
7. B
8. C
9. D
10. C
11. B
12. A
13. C
14. D
15. C
16. B
17. A
18. C
19. B
20. B

21. D
22. C
23. A
24. D
25. D
26. C
27. A
28. B
29. C
30. D

### **TIEBREAKERS**

31. B
32. D
33. A
34. C
35. C
36. A